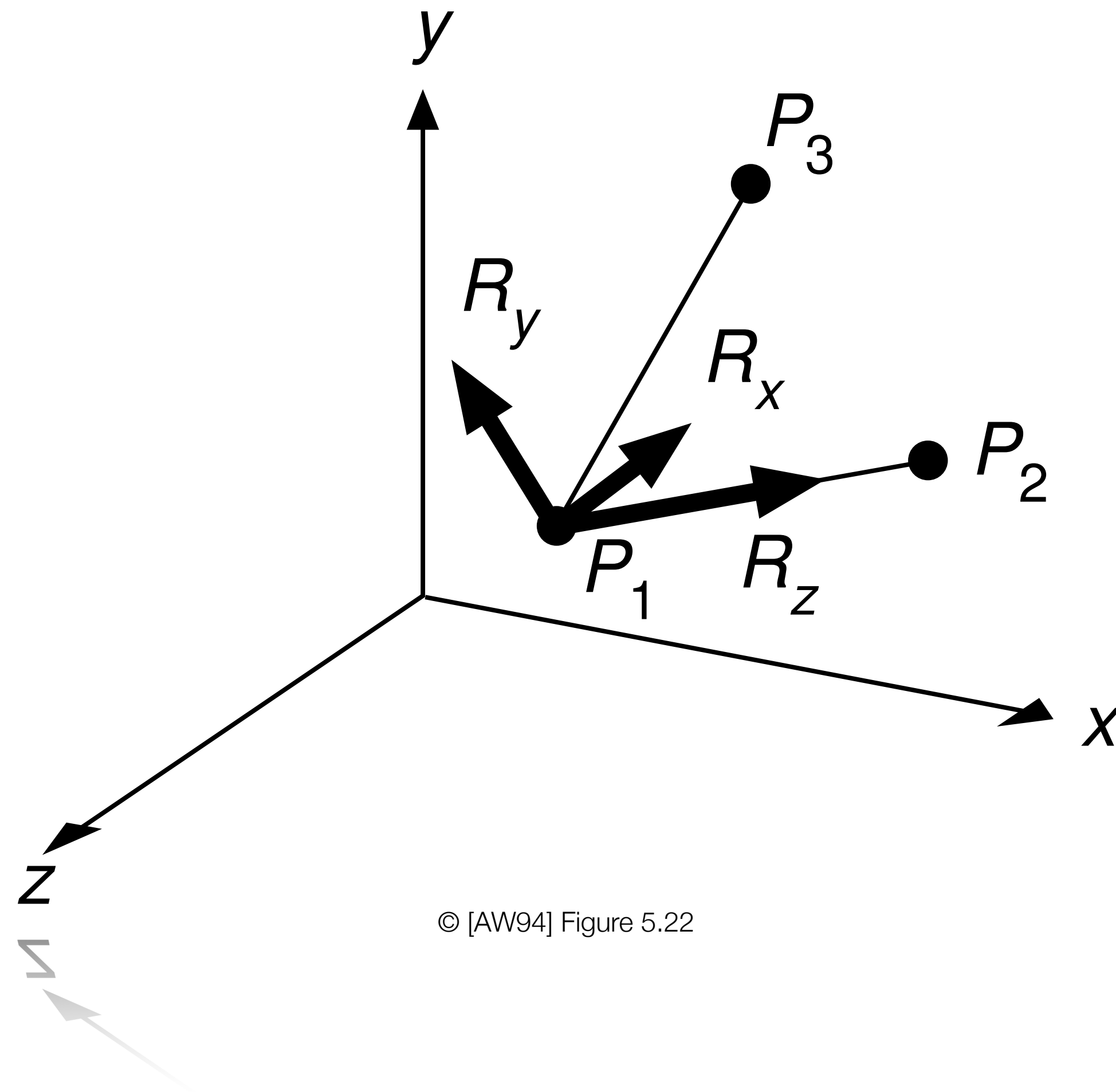


Scalars, Points and Vectors



© [AW94] Figure 5.22

Basic Geometric Objects

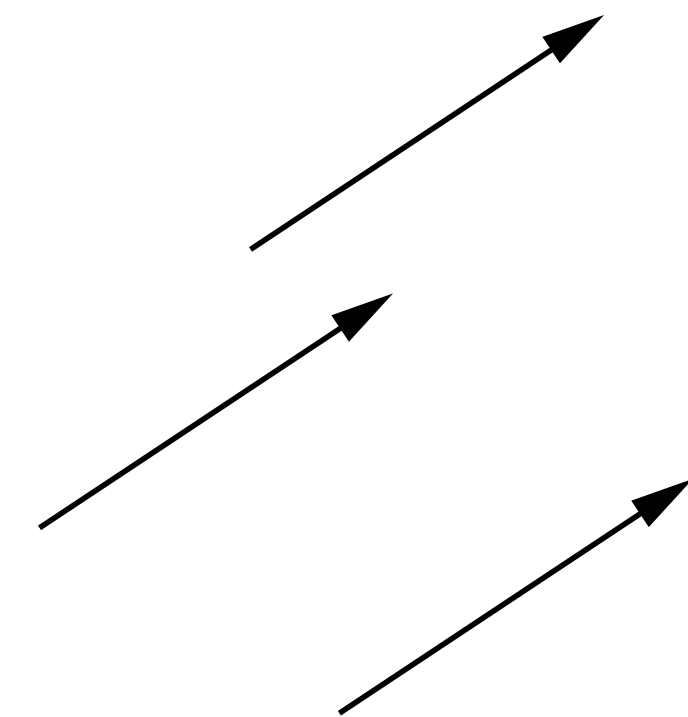
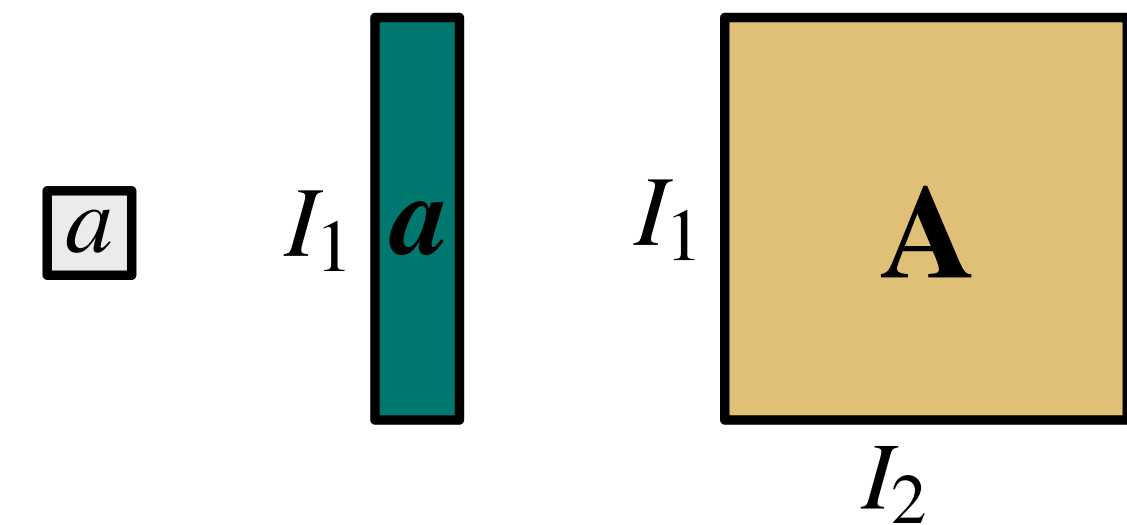
- Polygonal primitives and their relations can be described by few fundamental types:
scalars, points and vectors
- Geometric, mathematical and implementation view of these types may differ
 - ▶ geometric meaning, mathematical spaces, abstract data types

Scalars, Points or Vectors and Matrices

- Scalars, vectors, matrices
- Vectors denoting points and directions in space in N -dimensions
 - ▶ N -tuple of values, common column vector math notation

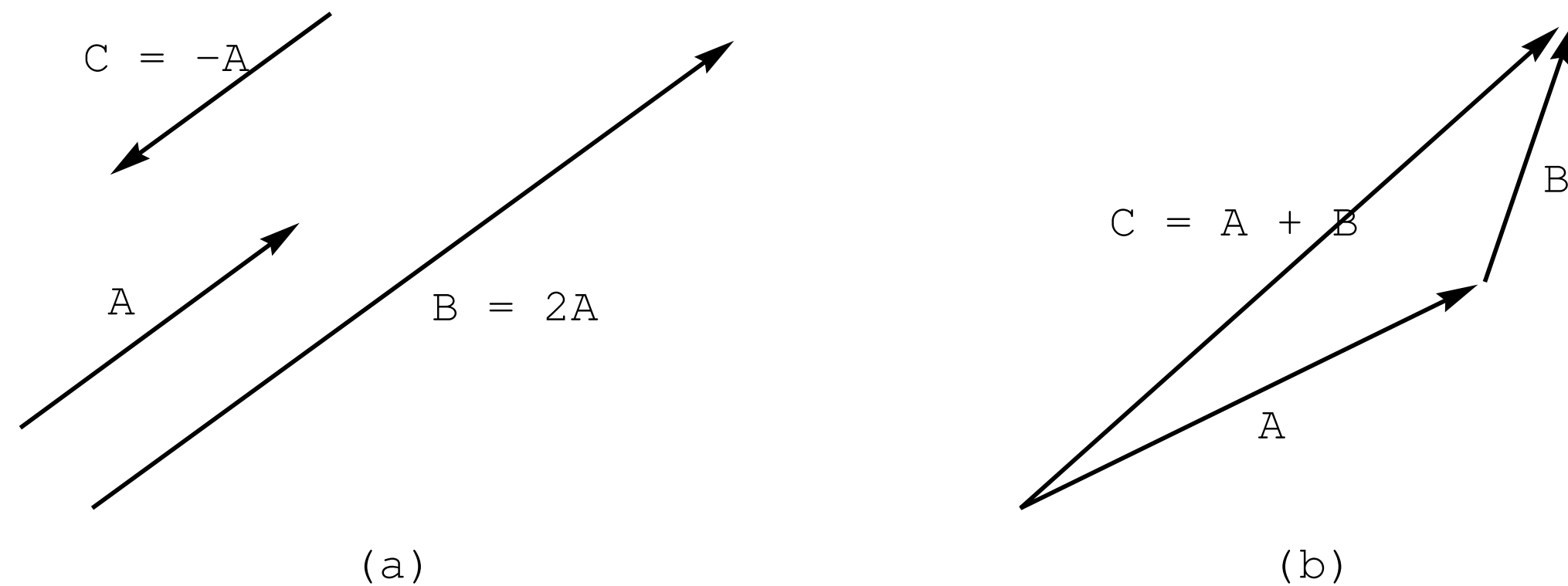
- vector $\mathbf{a} = \begin{pmatrix} a_1 \\ \vdots \\ a_N \end{pmatrix}$ commonly stored as 1D array $\mathbf{a} = [a_1, a_2, \dots, a_N]$

- A vector is a quantity with orientation and magnitude
 - ▶ e.g. velocity or force in physics
- Multidimensional data arrays for 2D matrices

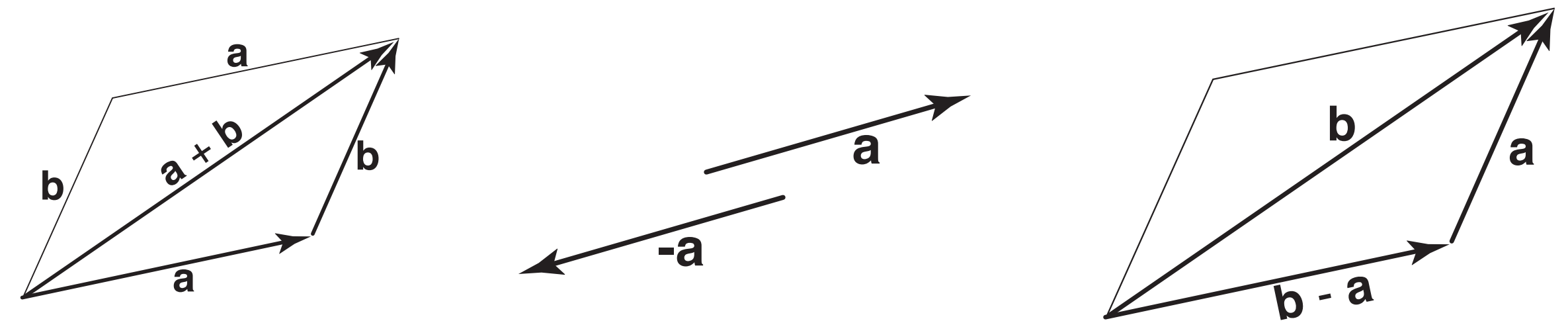


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Vector Addition and Subtraction



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© [S02] Figures 2.11, 2.13 and 2.14

- Vector addition is diagonal of parallelogram
 - ▶ *head-to-tail* rule of placing vectors
- Scalar multiplication and component-wise vector addition
 - ▶ scalar multiplication $\mathbf{b} = s \cdot \mathbf{a}$ is equivalent to adding $\mathbf{a}$ together s times

$$\mathbf{a} + \mathbf{b} = \begin{pmatrix} a_1 + b_1 \\ \vdots \\ a_N + b_N \end{pmatrix} \quad \mathbf{b} = s \cdot \mathbf{a} = \begin{pmatrix} s \cdot a_1 \\ \vdots \\ s \cdot a_N \end{pmatrix}$$

- Subtraction is equivalent to adding a negative vector
 - ▶ vector negation is inversion of direction
- Vectors are equal if of same length and direction
 - ▶ subtract to a zero-vector

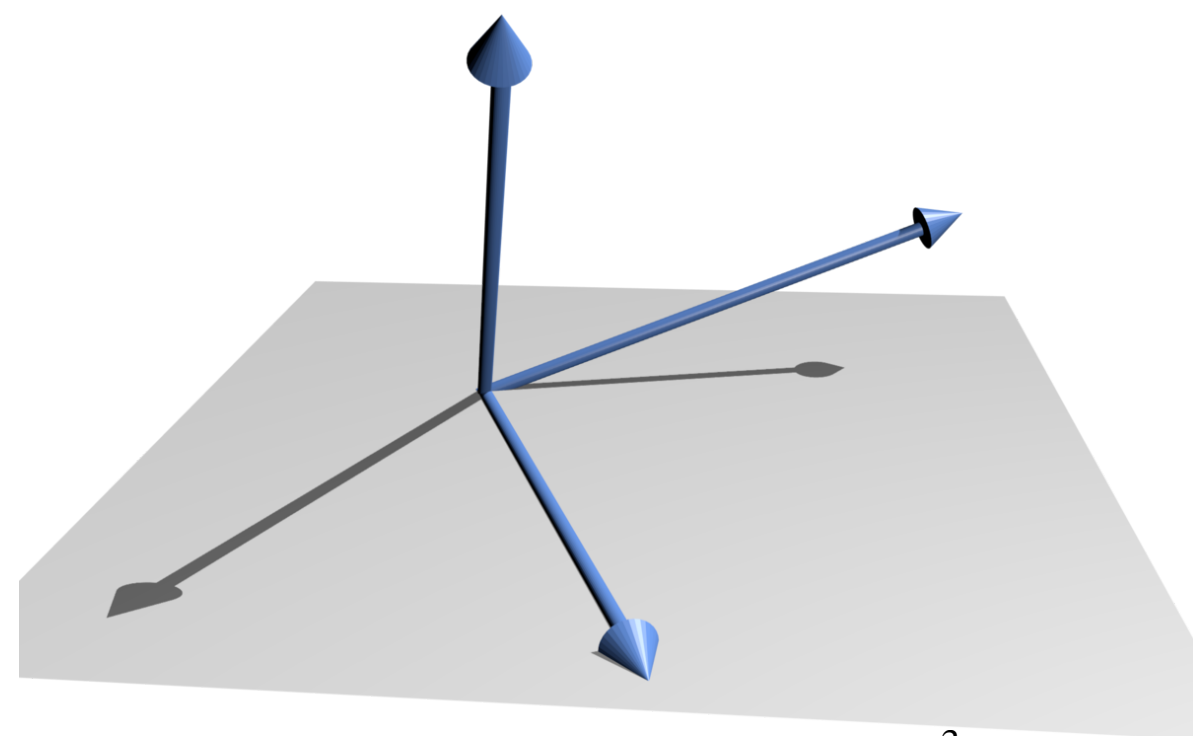
$$\mathbf{a} - \mathbf{b} = \begin{pmatrix} a_1 - b_1 \\ \vdots \\ a_N - b_N \end{pmatrix} = \mathbf{a} + (-1 \cdot \mathbf{b})$$

Vector Space

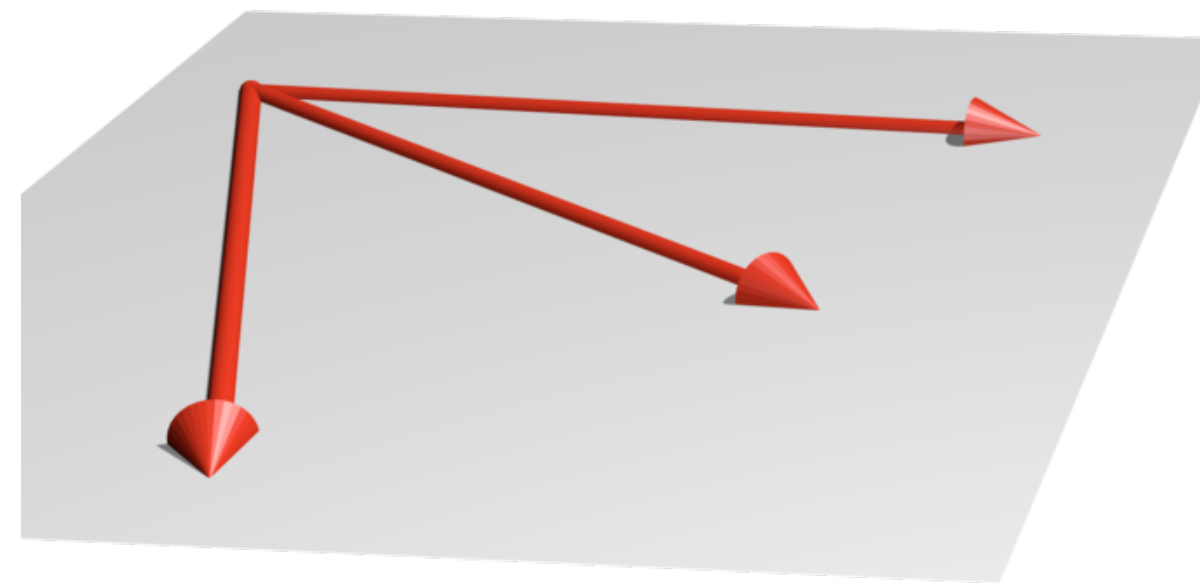
- In vector space V , addition and multiplication satisfy certain mathematical rules
 - closed: $\mathbf{u} + \mathbf{v} \in V, \forall \mathbf{u}, \mathbf{v} \in V$
 - commutative: $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
 - associative: $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$
 - distributive: $\alpha(\mathbf{u} + \mathbf{v}) = \alpha\mathbf{u} + \alpha\mathbf{v}$, $(\alpha + \beta)\mathbf{u} = \alpha\mathbf{u} + \beta\mathbf{u}$ for scalars α, β
 - zero vector: $\mathbf{u} + \mathbf{0} = \mathbf{u}$
 - additive inverse: $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$
- A vector space contains vectors as elements only !

Linear Independence

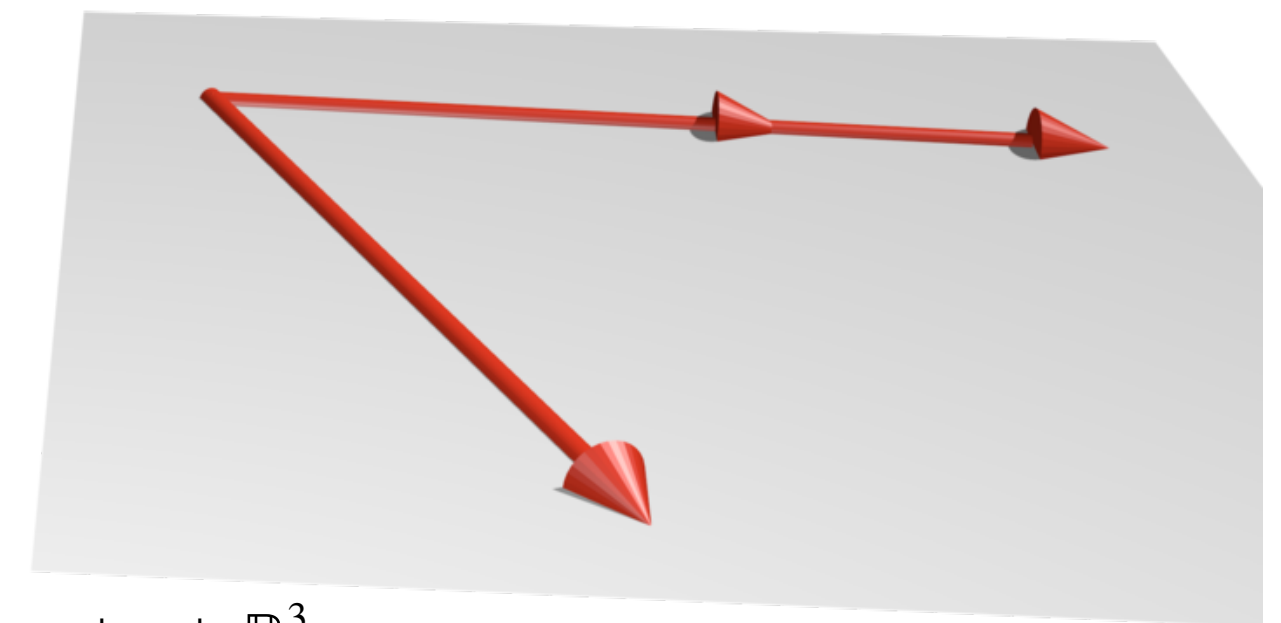
- A linear combination of N vectors $\mathbf{u}_1 \dots \mathbf{u}_N$ is defined as $\mathbf{u} = \alpha_1 \cdot \mathbf{u}_1 + \alpha_2 \cdot \mathbf{u}_2 \dots + \alpha_N \cdot \mathbf{u}_N$
 - ▶ linearly independent if $\alpha_1 \cdot \mathbf{u}_1 + \alpha_2 \cdot \mathbf{u}_2 \dots + \alpha_N \cdot \mathbf{u}_N = \mathbf{0}$ only for all $\alpha_i = 0$
 - ▶ the dimension of a vector space is defined by the largest number of linearly independent vectors
- N linearly independent vectors $\mathbf{v}_1 \dots \mathbf{v}_N$ form a basis of an N -dimensional vector space $\mathbf{V}$
 - ▶ $\forall \mathbf{v} \in \mathbf{V}, \exists \beta_1 \dots \beta_N \in \mathbb{R}$ such that $\mathbf{v} = \beta_1 \cdot \mathbf{v}_1 + \beta_2 \cdot \mathbf{v}_2 \dots + \beta_N \cdot \mathbf{v}_N$
 - ▶ orthonormal basis iff each vector is of unit length and is orthogonal to all others



linearly independent vectors in $\mathbb{R}^3$

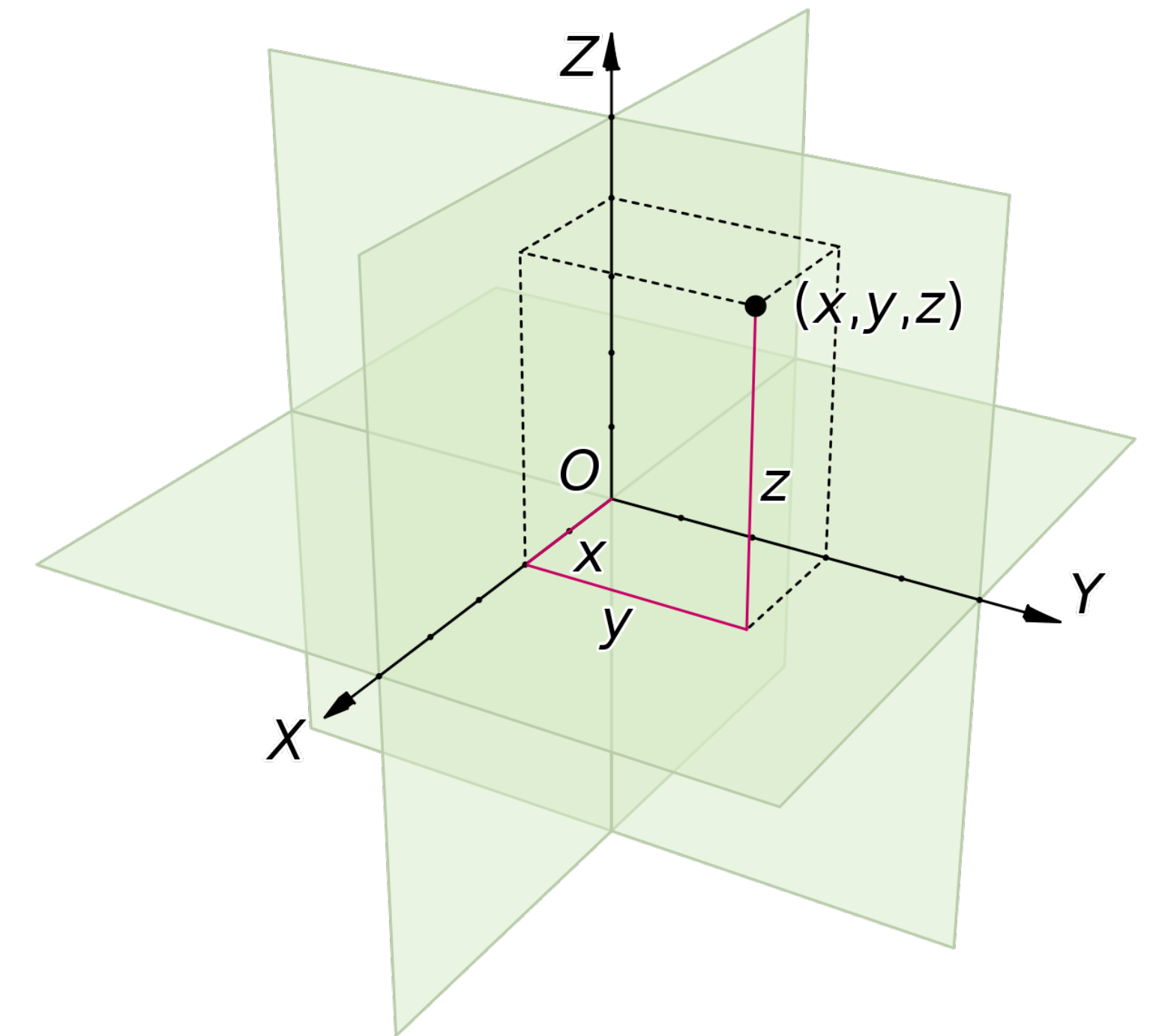


linearly dependent vectors in a plane in $\mathbb{R}^3$



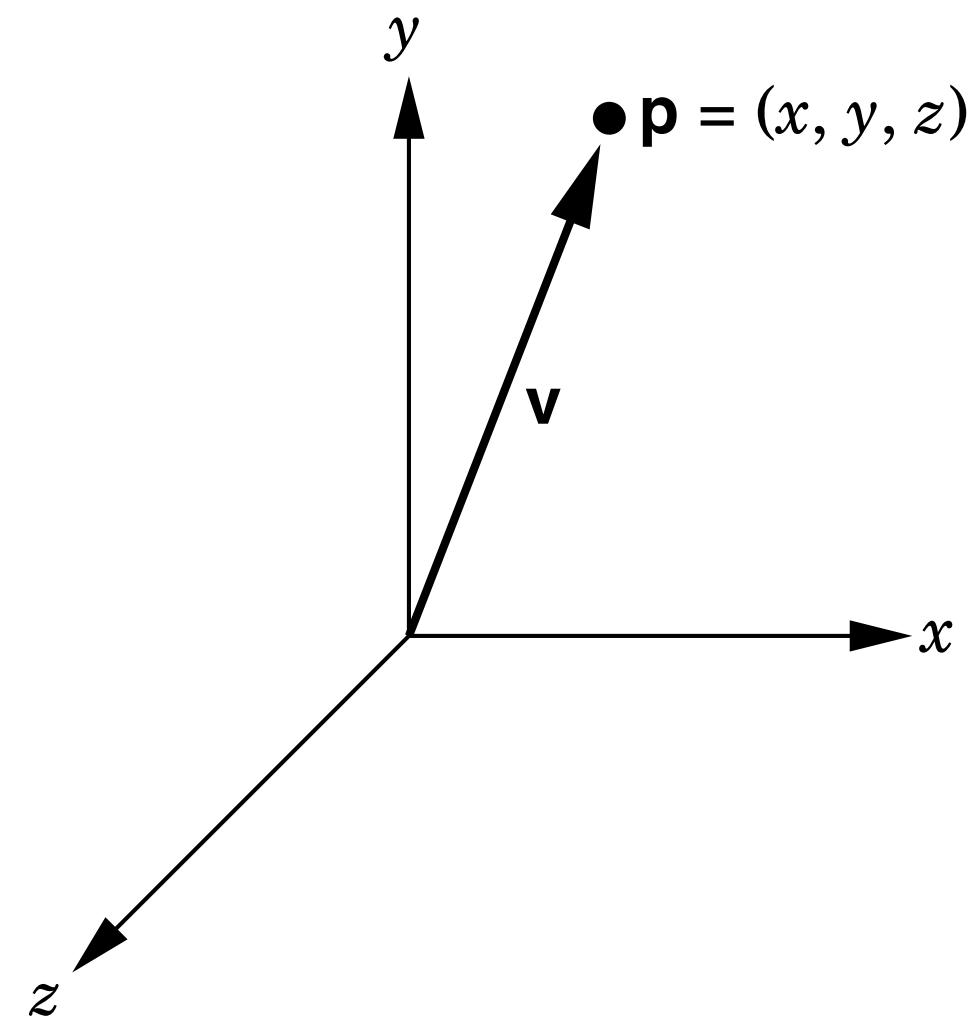
Cartesian Coordinate System

- A **Cartesian** coordinate system is defined by a set of orthonormal basis vectors
 - ▶ this basis defines the Cartesian coordinates of vectors
- We Given by the standard Cartesian N -dimensional basis vectors $\mathbf{v}_{i=1,\dots,N} = (\delta_{i1}, \dots, \delta_{ij}, \dots, \delta_{iN})^T$ with components δ_{ij}
 - ▶ Kronecker delta δ_{ij} is 0 for $i \neq j$ and 1 for $i = j$
 - ▶ for 3D we get the expected principal axis vectors:
 $\mathbf{v}_1 = (1,0,0)$, $\mathbf{v}_2 = (0,1,0)$ and $\mathbf{v}_3 = (0,0,1)$
 - ▶ vector $\mathbf{u}$ as column vector or one-colum matrix $\mathbf{u}^T = [u_x, u_y, u_z]$



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Points



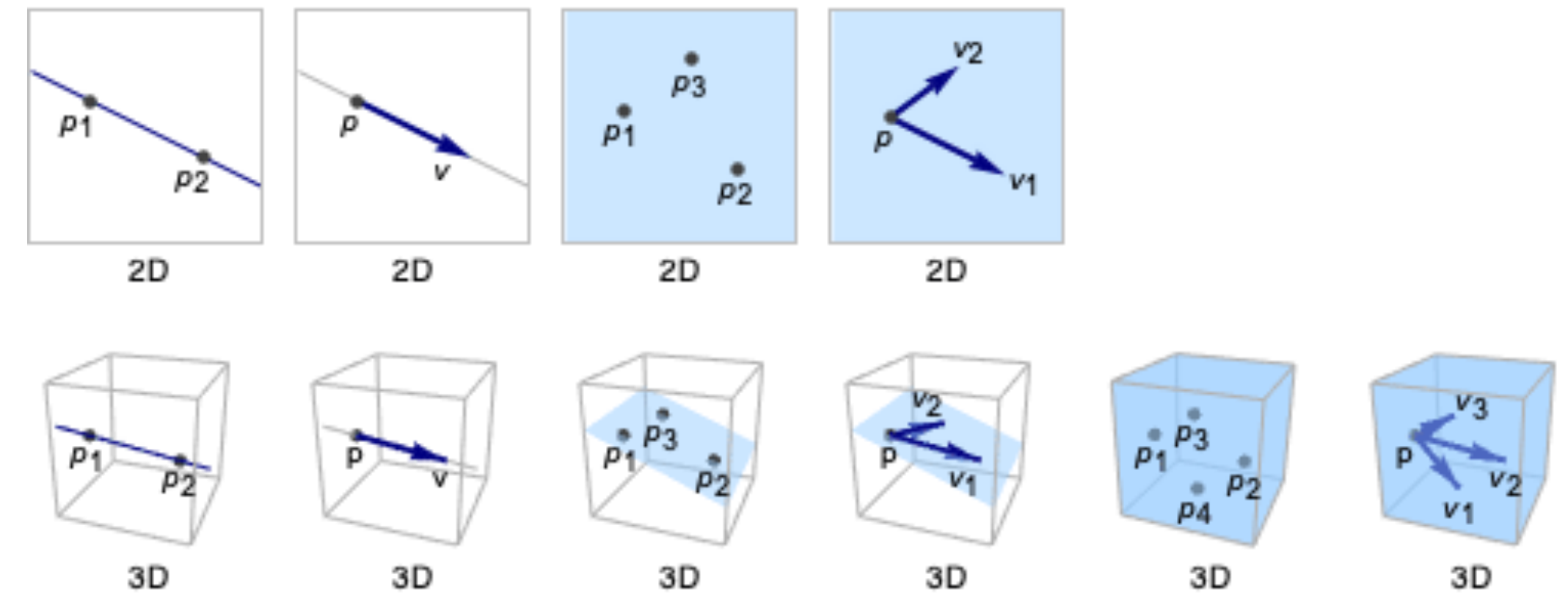
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- Points are locations in space and different from vectors
 - vectors lack position in space
- Points depend on an origin
 - defined as position vectors from the origin

Affine Space



© [A00] Figures 4.5 and 4.8



Source: <https://reference.wolfram.com/language/ref/AffineSpace.html>

- An affine space contains points in addition to scalars and vectors
 - ▶ operations: point-vector addition and point-point subtraction
 - add a vector to a point to get a new point
 - subtract two points to get the vector in between them
 - ▶ **no addition** of points or **multiplication** of points
 - cannot scale a point by scalar multiplication
- A line $L(t): P = Q + t \cdot (R - Q)$ is an affine space
 - ▶ linear interpolation $(1-t) \cdot Q + t \cdot R$ (line segment $t \in [0,1]$)
- A line, a plane or a volume in 3D represents an affine (sub-) space

Vectors in Graphics

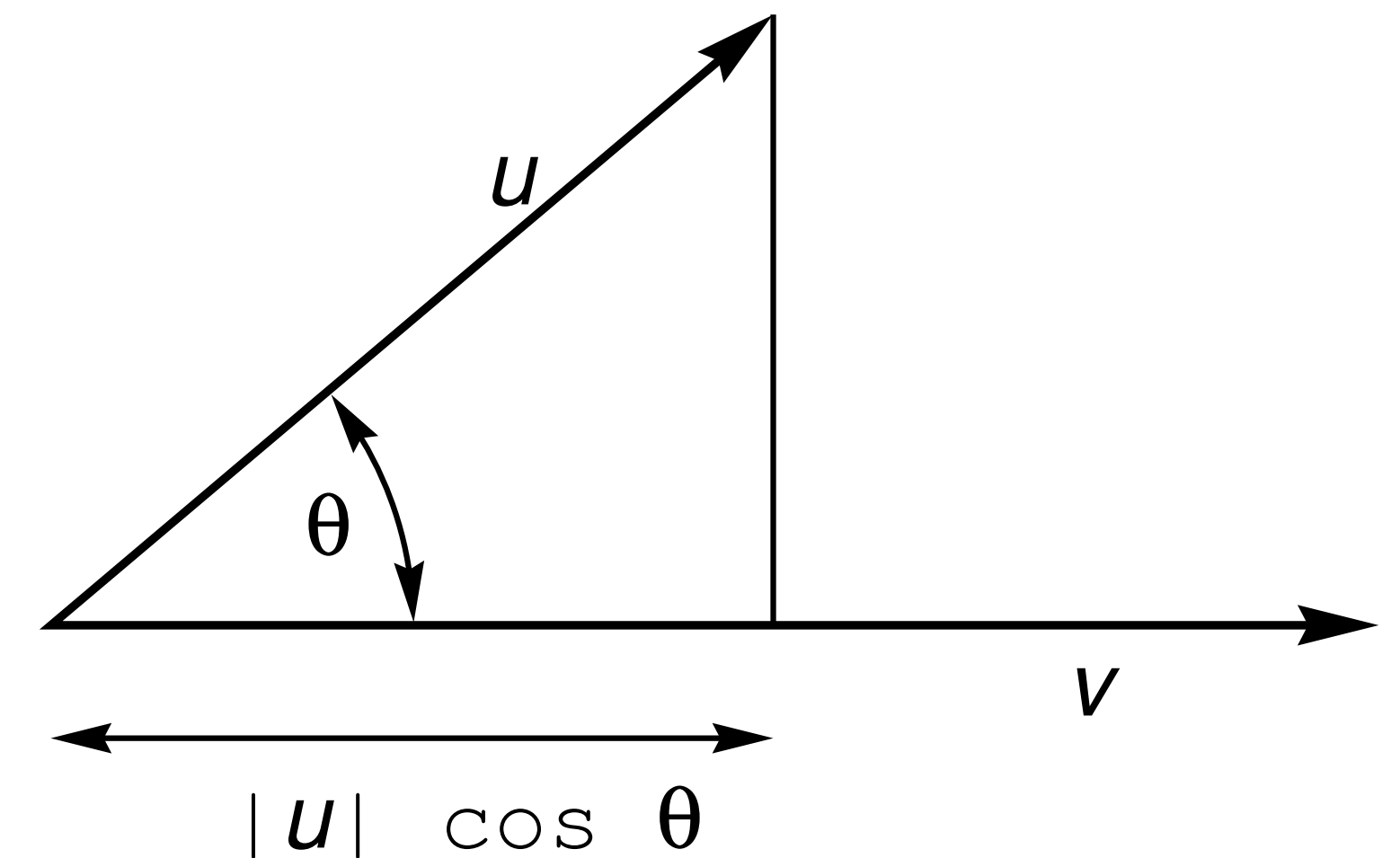
- Vectors are used to define the orientation of surfaces
 - ▶ normal of planes, polygons, triangles
- Relations between vectors are used to determine the interaction of light with surfaces
- Vectors are also (mis-) used to specify point locations in space as position vectors
 - ▶ displacement to point location from the origin

Definition of the Dot-Product

- Given two N -D vectors $\mathbf{u}$ and $\mathbf{v}$ the inner- or dot-product ‘ $\cdot$ ’ is defined as $\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + \dots + u_n v_n$
 - ▶ operation on two vectors with a scalar result
- The dot-product is
 - ▶ symmetric: $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$
 - ▶ non-degenerate: $\mathbf{v} \cdot \mathbf{v} = 0 \Leftrightarrow \mathbf{v} = \mathbf{0}$
 - ▶ bilinear: $\mathbf{v} \cdot (\mathbf{u} + \alpha \mathbf{w}) = \mathbf{v} \cdot \mathbf{u} + \alpha (\mathbf{v} \cdot \mathbf{w})$

Geometric View of the Dot-Product

- Relation to angle between vectors: $\mathbf{u} \cdot \mathbf{v} = \cos \theta \cdot |\mathbf{u}| \cdot |\mathbf{v}|$
 - ▶ $\mathbf{u}, \mathbf{v}$ point to opposite half-spaces if negative
 - ▶ vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal if $\mathbf{u} \cdot \mathbf{v} = 0$
- Orthogonal projection of $\mathbf{u}$ onto $\mathbf{v}$ given by: $|\mathbf{u}| \cdot \cos \theta = \mathbf{u} \cdot \mathbf{v} / |\mathbf{v}|$
 - ▶ for **unit** length vector $\mathbf{v}$, it is simply $\mathbf{u} \cdot \mathbf{v}$
- Square length of vector: $|\mathbf{u}|^2 = \mathbf{u} \cdot \mathbf{u}$

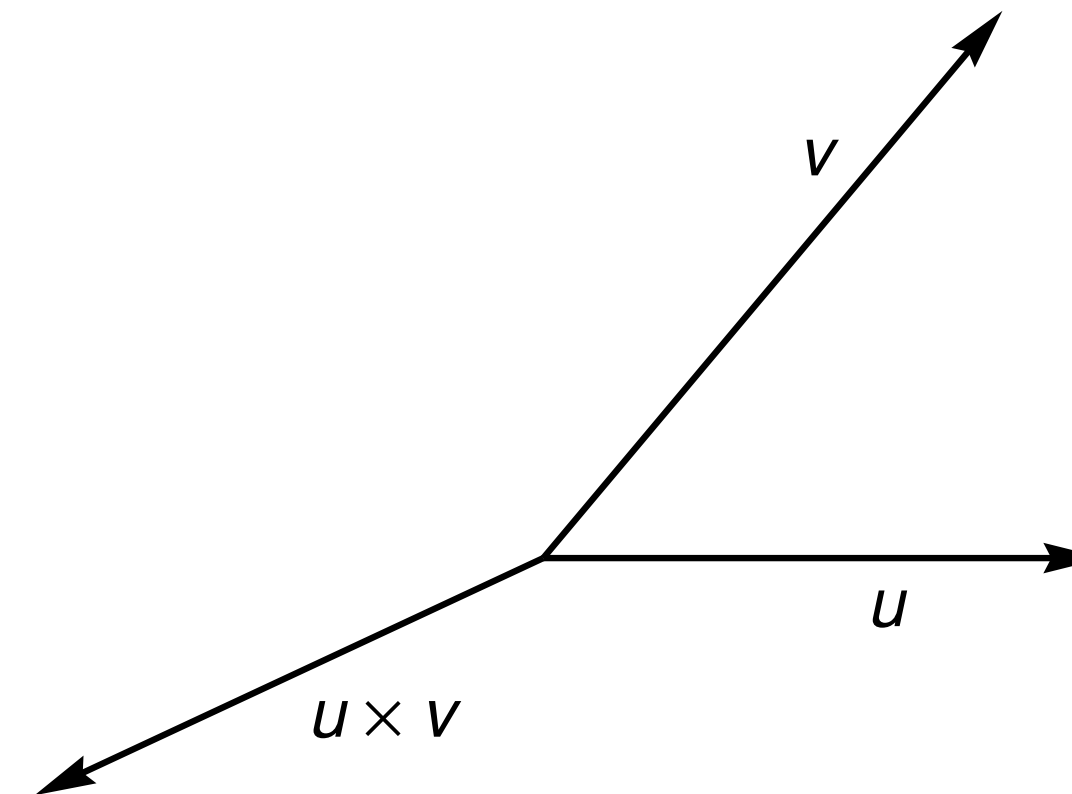


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Definition of the Cross-Product

- Given two 3D vectors u and v their cross-product $w = u \times v$ is defined as

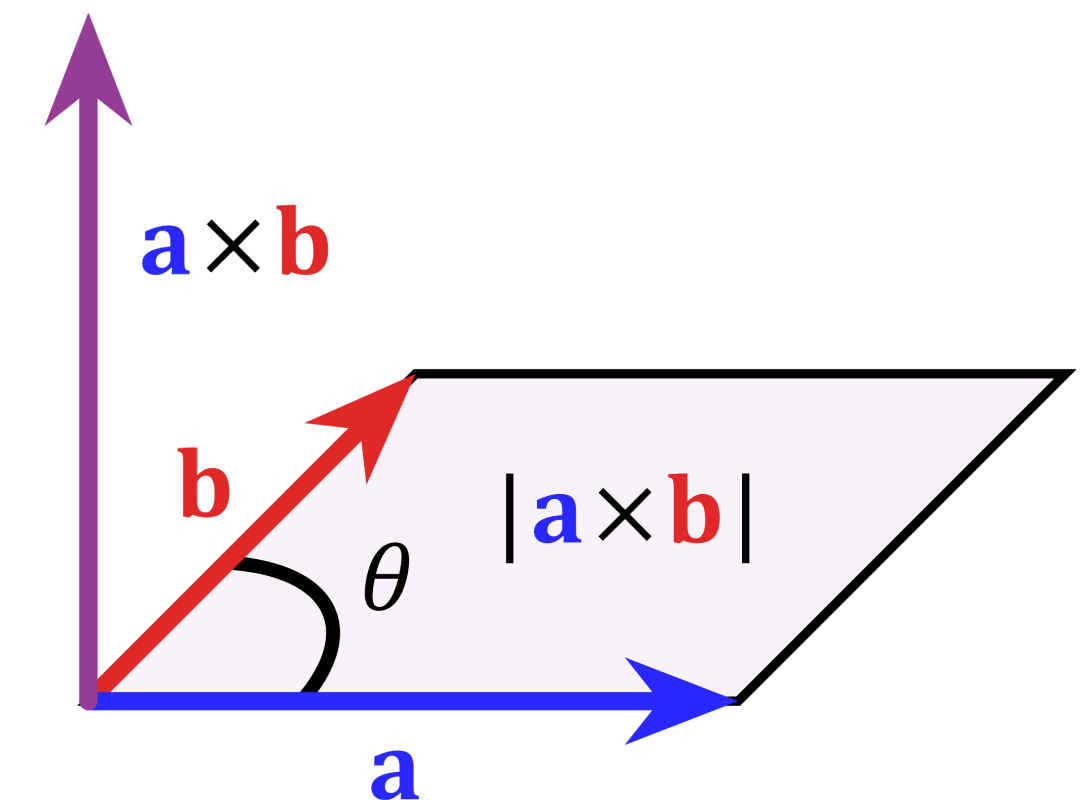
$$\begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix} = \begin{pmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{pmatrix}$$



© [A00] Figure 4.12

Geometric View of the Cross-Product

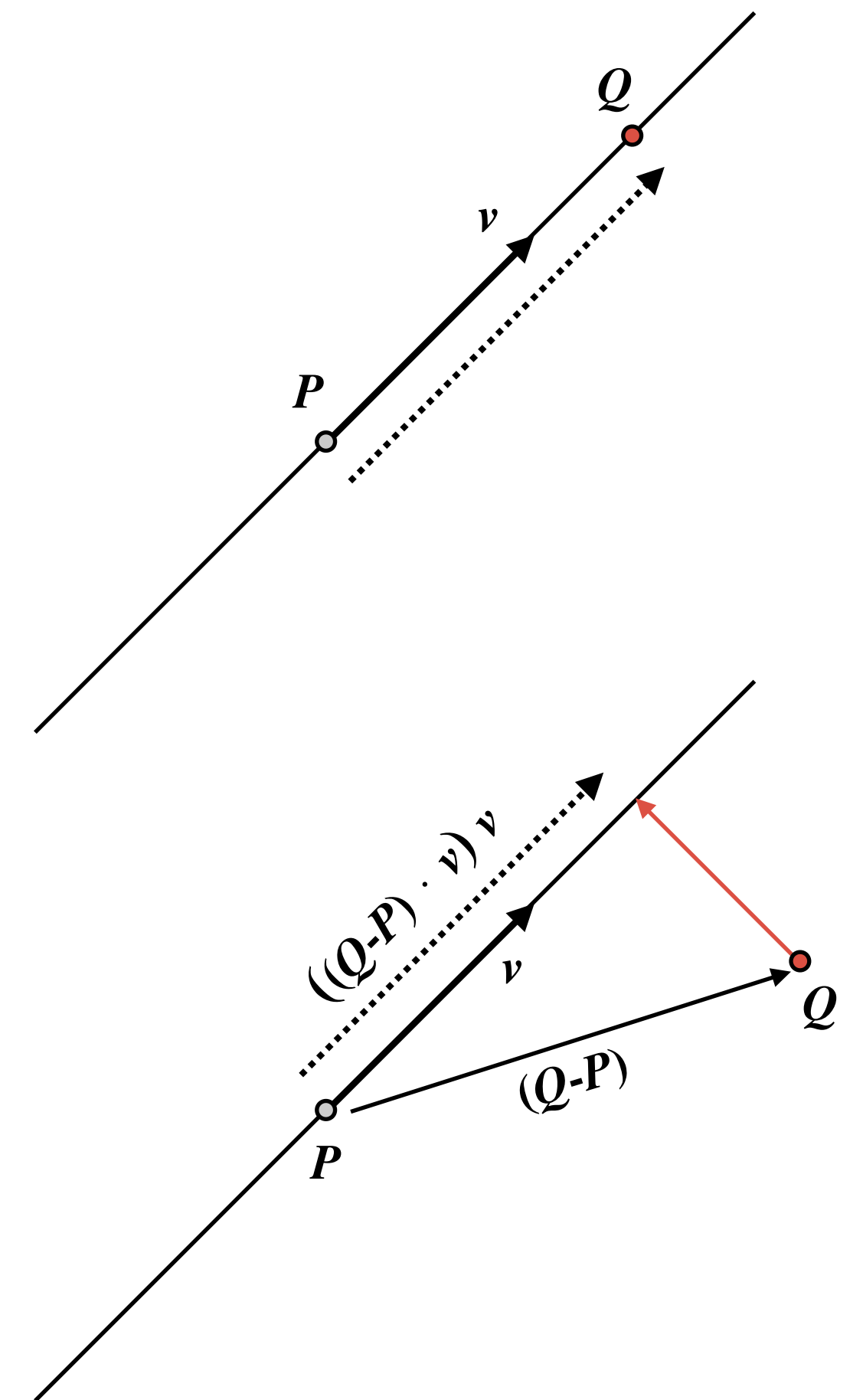
- The cross-product $\mathbf{w} = \mathbf{u} \times \mathbf{v}$ is perpendicular to $\mathbf{u}$ and $\mathbf{v}$
 - ▶ defines a right-handed vector system
- Normal of triangle (P, Q, R) is $\|(Q-P) \times (R-P)\|$
 - ▶ $\|\mathbf{v}\| = \mathbf{v} / |\mathbf{v}|$ denotes normalization to unit length
- Relation to angle between vectors: $|\mathbf{u} \times \mathbf{v}| = |\sin \theta| \cdot |\mathbf{u}| \cdot |\mathbf{v}|$
 - ▶ vectors $\mathbf{u}$ and $\mathbf{v}$ are collinear if $\mathbf{u} \times \mathbf{v} = \mathbf{0}$
- Length $|\mathbf{u} \times \mathbf{v}|$ is area of parallelogram spanned by $\mathbf{u}$ and $\mathbf{v}$
 - ▶ area of triangle (P, Q, R) is given by $1/2 \cdot \|(Q-P) \times (R-P)\|$



Source: <https://stackoverflow.com/questions/32146434/determinant-method-vs-cross-product-area-of-a-triangle>

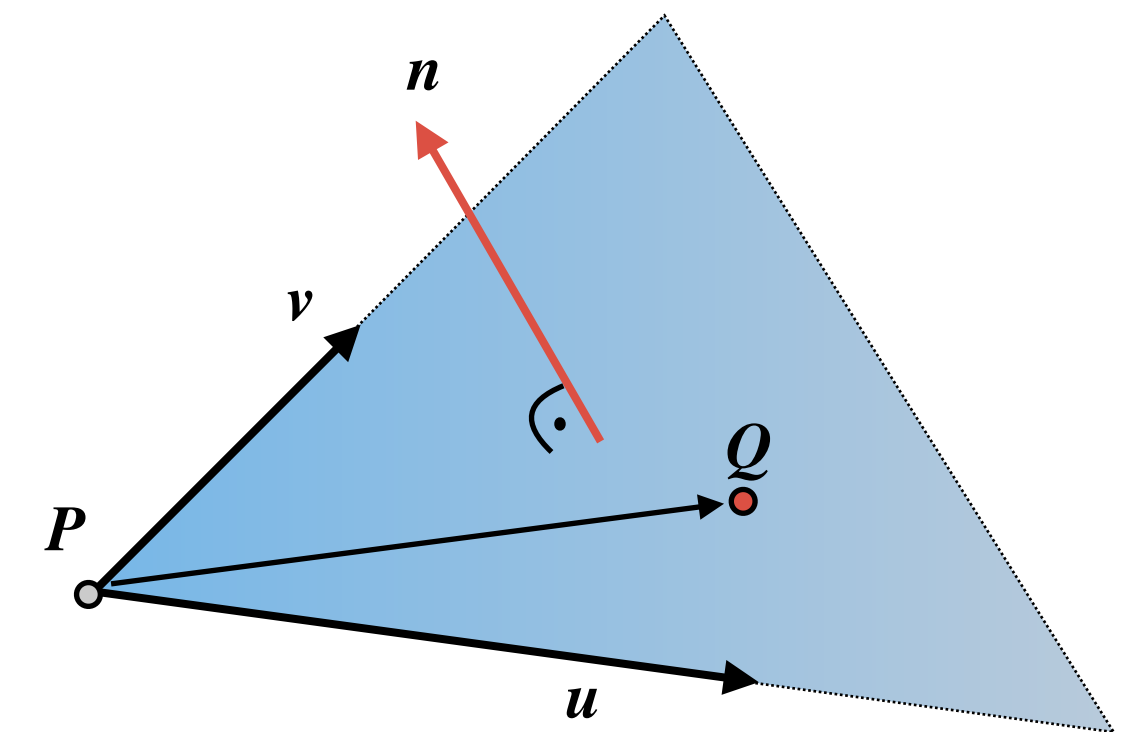
Lines

- A line can be expressed in parametric form as $L(t) = P + t \cdot v$
 - ▶ given a point P on the line and the direction v of the line
- The dot-product can describe a line implicitly as $Q \in L$:
$$(Q-P) \cdot v == |v| \cdot |Q-P|$$
 - ▶ Q on line L if $\theta = 0.0$, angle between $Q-P$ and v , and thus $\cos \theta = 1.0$
- Distance of Q to line can be described by $d = | (Q-P) - ((Q-P) \cdot v) v |$
 - ▶ using normalized unit-length line direction $|v| = 1.0$



Planes

- A plane can be expressed in parametric form as $M(s,t) = P + s \cdot \mathbf{u} + t \cdot \mathbf{v}$
 - ▶ given a point P and two vectors $\mathbf{u}$ and $\mathbf{v}$ in the plane
 - ▶ affine sum and linear combination of vectors – **affine space**
- The cross-product can describe a plane implicitly as $Q \in M$:
$$| \|Q-P\| \times \mathbf{n} | == 1$$
 - ▶ normalized vector $Q-P$ orthogonal to normal $\mathbf{n}$, thus $\sin \theta = 1.0$
 - ▶ given a point in the plane P and unit-length plane normal $\mathbf{n}$
 - ▶ but simple(r) test using dot-product $(Q-P) \cdot \mathbf{n} = 0$, for $\cos \theta = 0.0$
- Generate normal and orthogonal coordinate axis by cross-products



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